

SPLUNK SOC INVESTIGATION REPORT

Suspicious Login and PowerShell Activity — CORP-WS-042

Case ID	SOC-2024-0315-001
Title	Suspicious Login and PowerShell Activity
Date	2024-03-15
Affected Host	CORP-WS-042
Affected User	jsmith
Tool Used	Splunk Enterprise (index: soc_lab)
Severity	HIGH
Status	Open — Under Investigation

Executive Summary

On 2024-03-15 at 14:32 UTC, a SIEM alert fired for suspicious PowerShell execution on host CORP-WS-042, associated with user jsmith. Investigation in Splunk identified a pattern of failed login attempts followed by a successful login, an obfuscated PowerShell command, and outbound network connections to an external IP address. This sequence is consistent with unauthorized access followed by command execution and an attempt at external communication. Severity is assessed as HIGH, and the host is recommended for escalation and further containment.

Investigation Methodology

Log data was ingested into Splunk under index soc_lab from three sources: Windows Security logs, PowerShell logs, and network connection logs. The investigation followed a structured workflow:

1. Reviewed login activity for the affected user and host
2. Analyzed PowerShell execution for suspicious command-line flags
3. Reviewed outbound network connections tied to the suspicious process
4. Reconstructed a timeline correlating all findings
5. Built a four-panel SOC dashboard to consolidate evidence
6. Extracted indicators of compromise and mapped activity to MITRE ATT&CK
7. Assigned a severity rating and recommended next steps

Findings: Login Activity

Host: CORP-WS-042 | User: jsmith

Windows Security logs showed multiple failed login attempts for jsmith, immediately followed by a successful login. This pattern is suspicious and is consistent with brute-force or credential-stuffing activity that ultimately succeeded.

Findings: PowerShell Execution

Event Time: 2024-03-15 14:32:17 UTC | Host: CORP-WS-042 | User: jsmith | Process: powershell.exe

Suspicious Flags Identified

- nop Bypasses the PowerShell profile
- w hidden Hides the PowerShell console window
- enc Base64-encodes the command (obfuscation)

These flags are commonly associated with malicious PowerShell activity, since they can hide the command window, bypass normal profile loading, and obscure the true command through encoding.

MITRE Mapping

- T1059.001 — PowerShell
- T1027 — Obfuscated Files or Information

Findings: Network Connections

Source Host: CORP-WS-042 (10.10.5.42) | Destination IP: 185.220.101.47

- Connection 1 — Destination Port 443, via powershell.exe
- Connection 2 — Destination Port 80, via powershell.exe

Outbound connections from CORP-WS-042 to the external IP 185.220.101.47 occurred shortly after the suspicious PowerShell execution and were associated with powershell.exe. PowerShell should not normally generate direct outbound web connections during regular user activity, which supports the conclusion that the encoded command attempted to communicate with an external server.

MITRE Mapping

- T1071.001 — Web Protocols
- T1105 — Ingress Tool Transfer

Incident Timeline

- 14:21:05–14:21:22 UTC – Multiple failed login attempts for jsmith (Event ID 4625)
- 14:22:01 UTC – Successful login for jsmith (Event ID 4624)
- 14:32:17 UTC – Suspicious PowerShell command executed (-nop, -w hidden, -enc)
- 14:32:19 UTC – Outbound connection to 185.220.101.47 on port 443
- 14:32:21 UTC – Outbound connection to 185.220.101.47 on port 80

The timeline shows failed login attempts followed by a successful login, suspicious encoded PowerShell execution, and outbound network activity — a sequence suggesting possible unauthorized access followed by command execution and external communication.

SOC Dashboard

Dashboard Name: SOC Investigation — CORP-WS-042

A four-panel Splunk dashboard was built to organize investigation evidence in one place:

- Panel 1 — Login Events Timeline: failed and successful logins over time (Event IDs 4625 / 4624)
- Panel 2 — PowerShell Execution Events: command-line activity in table form to surface the suspicious command
- Panel 3 — Outbound Network Connections: destination IPs, ports, and processes involved
- Panel 4 — Event Summary by Source: event counts from each ingested log source

Panels 2 and 3 were kept as tables rather than charts, since the raw command-line and connection details are more useful than visual summaries for this stage of the investigation.

Indicators of Compromise

Indicator Type	Value	Confidence
Host	CORP-WS-042	Confirmed
User Account	jsmith	Confirmed
Internal Source IP	10.10.5.42	Confirmed
External Destination IP	185.220.101.47	High
URL	http://185.220.101.47/payload.ps1	High
File / Script Name	payload.ps1	Medium
Process	powershell.exe	High
Command Flags	-nop, -w hidden, -enc	High

MITRE ATT&CK Mapping

Technique ID	Technique Name	Tactic
T1110	Brute Force	Initial Access
T1059.001	PowerShell	Execution
T1027	Obfuscated Files or Information	Defense Evasion
T1071.001	Web Protocols	Command and Control

Technique ID	Technique Name	Tactic
T1105	Ingress Tool Transfer	Command and Control

Together, these techniques indicate possible unauthorized access, obfuscated execution, and an attempt at external communication consistent with a command-and-control connection attempt.

Severity Assessment

Severity: **HIGH**

This incident is rated High because the investigation found multiple suspicious behaviors connected within a single timeline:

1. Multiple failed login attempts followed by a successful login
2. Suspicious PowerShell execution using -nop, -w hidden, and -enc
3. A decoded command that attempted to download a remote script
4. Outbound connections from powershell.exe to an external IP address
5. Evidence of possible command execution and external communication

This activity suggests possible unauthorized access followed by malicious PowerShell execution and outbound communication. The host should be escalated for containment and further investigation.